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Irradiation-induced creep in nanocrystalline Cu alloys

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ABSTRACT

Irradiation-induced creep in nanocrystalline Cu was simulated with the aim of analyzing the microscopic mechanism driving creep. The systems included various immiscible mixtures where the solute atom segregated at grain boundaries and led to grain size stabilization. The small grain size in the nanocrystals prevents the development of dislocation-based dynamics within the grains, and gives rise to alternative mechanisms that are based on grain-boundary plasticity. We show a correlation between observed creep rates and the climbing of dislocations at the grain boundaries. Due to the simple structure of Cu-based alloys, they can serve as model systems for investigating irradiation-induced creep. This establishes the basis for a mean-field model that can predict the creep compliance of a sample as a function of its structure and composition. The model reproduces recent experimental measurements.

1. Introduction

Structural materials used in an environment exposed to irradiation are prone to degradation of mechanical properties and other detrimental effects. Irradiated particles transmit kinetic energy to a lattice atom, inducing a supersaturation of point defects, which are only partly recombined. The remaining point defects may cluster, form interstitial loops, or interact with dislocations, allowing enhanced climbing and gliding. These, in turn, lead to dimensional instabilities and irradiation-induced creep (IIC) [1]. This mandates the development of new materials that are more resistant to irradiation.

One approach is to use nanocrystalline metals [2–4], in which interactions between dislocations and grain boundary (GB) interfaces lead to a dislocation starvation effect [5,6], thus preventing the possibility of IIC due to the motion of dislocations within the interior of the grains. In general, creep in nanocrystals (NCs) is commonly attributed to plastic processes originating in the GBs themselves, such as Coble creep [7,8], GB sliding, and emission and absorption of dislocations at the GBs [9]. In addition, stress-driven GB migration can lead to grain coarsening and ultimately to dislocation creep within the bulk [10–12], or to shear deformation [13–17].

Some of these deformation mechanisms can be mitigated by introducing immiscible solute atoms into the NC material [12,18,19]. These can stabilize the GBs either energetically or by limiting GB mobility, by forming precipitates leading to Zener pinning [20]. In such alloys, the creep rate depends on the interaction between the solute and matrix atoms and on their concentrations [5,21,22].

In the case of irradiated NCs, due to the large volume fraction of the GBs, point defects that did not recombine diffuse to the GBs without significant dependence on external stress [1]. Thus, creep occurs because of the interaction of these point defects with sinks and sources at the GBs, rather than the anisotropic flow of defects in the applied stress field [2,3]. IIC was studied in Cu and Cu-W, which are convenient model systems due to their simple structure, as well as being of interest in and of themselves. It was found that below a temperature of 400 K, IIC is driven by stress, with a negligible thermal component [2,3]. This is in line with previous studies, which showed that thermal creep mechanisms such as Nabarro–Herring and Coble creep are unaffected by irradiation [1,3,8].

This study aims to characterize the effect of the alloy composition and stress on IIC in fcc Cu-X NC alloys, where X is an immiscible solute atom, focusing on the nonthermal regime. This is done by applying molecular dynamics (MD) to examine irradiation-induced creep compliance (see Section 3.1), as well as the mesoscopic and microscopic mechanisms leading to IIC, in a range of simulated systems. The findings are then compared to the experimental results in such systems [3,4].

2. Methods

To study IIC via simulations, MD was applied to track the response of a NC sample to the introduction of defects in the GBs, in line with the work done in Ref. [2]. It is assumed that the flow of vacancies and

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interstitials to sinks does not introduce significant diffusional creep [1, 2]. All MD simulations were performed using large-scale atomic molecular massively parallel simulators (LAMMPS) [23]. A cubic superlattice of fcc Cu was created, with a length of 22 nm, containing 9.5×10^5 atoms. The superlattice was divided into grains of approximately equal size by choosing points within the superlattice, and then applying Voronoi tessellation to construct grains around them. Seventeen points were chosen, resulting in an average grain diameter of 10 nm. Each grain was oriented randomly, according to the distribution described in Ref. [2]. The sample was then relaxed by quenching to a temperature of 300 K and zero external pressure.

After the NC lattice was produced, solute atoms were added using the embedded atom method potentials described in Refs. [24–26] and the LAMMPS gcmc command [27], which combines a Monte-Carlo-based insertion of solute atoms from an imaginary ideal gas reservoir with a subsequent MD relaxation. Using this method, the chemical potential was varied between 1.2 and 3.8 eV to form a system with solute concentrations between 0.4% and 5.8%, correspondingly. After the desired concentration of solute was achieved, the samples were relaxed into a steady state using the LAMMPS nve command under periodic boundary conditions. Twenty-one samples of fcc NC Cu-X were thus created, where X is either Nb, V, or Ta, varying only by the identity and concentration of the solute atoms, see Fig. 1. We used these three solute atom types as they all have large size mismatches and high immiscibility, while their specific values and the potentials used to describe their interatomic interaction differ. Thus, the result of the analysis can be generalized by comparing their relative effects. In all three cases, the high immiscibility of the solute atoms within the grains led to GB segregation.

After the formation of the metastable samples, target stresses were applied along one direction, which we define as the z direction, by simulating the system in a nonisotropic npt ensemble. The target stress was constant at 3, 4, 5 or 6 kBar, and the temperature was set at 300 K. The simulation was run until no significant variation of the energy was observed over a period of 320 ps.

Following this initial stabilization, to simulate the interaction of radiation-induced defects with the GBs, point defects were introduced at random locations within the GBs. This was done by removing and reinserting 70 atoms at the GBs, followed by a short relaxation of the system, as in Ref. [2]. The dynamic response of the system was assumed to be due to the interaction of the defects with the GBs following the defect introduction. This was simulated using an npt ensemble for a period of 80 ps, and repeated over at least 1,200 cycles. We validated that the results were insensitive to the rate of defect introduction and the initial relaxation and setup, using the same methods as in Ref. [2].

The early cycles of defect introduction led to initial stage creep, which included elastic deformation and volume change. Once the elastic deformation came to an end, a plastic stage began. The creep rate at the plastic stage was constant.

3. Results

3.1. Dependence of the IIC rate upon the solute material and concentration

The accumulated amount of point defects reaching the GBs increases linearly with the intensity of irradiation. The intensity is normally measured as the number of displacements per atom (dpa). However, in the context of IIC due to GB dynamics, the relevant measure is the number of freely-migrating defects forming within the grain interior, reaching the GBs, and causing local relaxation. We thus define the effective dpa ϕ as the number of defects reaching the GBs normalized by the number of atoms in the GBs. To measure the susceptibility of the sample to deformation due to IIC, we use the creep compliance C, which is the derivative of the strain ϵ by the effective dpa normalized by the applied tensile stress σ, i.e., $C = d\epsilon/(\sigma d\phi)$. When the strain rate and the irradiation are constant in time, we have also

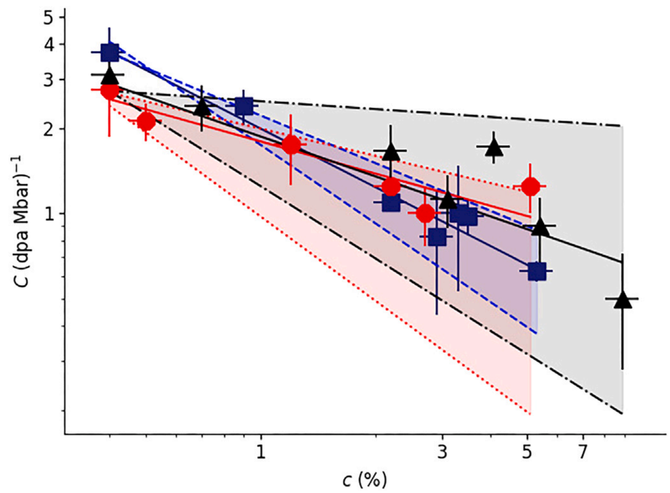

Fig. 1. Irradiation-induced creep compliance as a function of solute concentration at the GBs, for NC Cu-V (squares), Cu-Nb (circles), and Cu-Ta (triangles), under a stress of 4 kbar. The horizontal error bars represent the confidence level in the estimation of the number of atoms in the GBs, and the vertical error bars represent the confidence level of the strain-dpa fit used to calculate the creep compliance. Solid lines represent fits to Eq. (2) for each alloy, and the shaded bands show the confidence level of each fit.

$$C = \frac{d\dot{\epsilon}}{\sigma d\dot{\phi}}, \tag{1}$$

where $\dot{\epsilon}$ is the strain rate and $\dot{\phi}$ is the *effective dose rate*, given as the rate of the effective dpa. See Appendix A for more details regarding the calculation of C. The creep compliance showed a strong dependency on solute concentration, while being minimally affected by the added solute atom type. All solute atom types are larger than Cu and have a similar heat of mixing, see Fig. 1. The dependence of the creep compliance on the solute concentration was fitted to a power law

$$C = ac^d, \tag{2}$$

where a and d are fitting parameters. d was found to be 0.6 ± 0.2 for all solute types.

3.2. Driving mechanism of IIC

To investigate the mechanism by which point defects reaching the grain boundary drive IIC, we examined the displacements of atoms in the lattice between two time frames, 80 ps apart from each other. The distribution of displacements due to normal thermal motion is expected to be Gaussian, while plastic activity is connected to displacements that fall outside this distribution. As seen in Fig. 2 for one of the studied samples, the non-Gaussian displacements are concentrated at the GBs. Similar results were obtained for all the samples in this study, demonstrating that the IIC is driven by processes at the GBs, rather than by the emission of dislocations into the grain interior.

As described in Section 1, IIC driven by plastic processes at the GBs can be mediated by GB sliding [9] and stress-driven GB migration causing grain coarsening or shear deformation. [10–12,15]. In our simulations, GB sliding was shown to be insignificant by monitoring the relative displacement of atoms in adjacent grains, see Appendix B. This leaves grain coarsening or shear deformation, due to GB migration, as the probable mechanisms of IIC in our simulations. Thus, measuring the flux of atoms through the GBs provides an estimate of the extent of GB migration.

Throughout each simulation, atoms entered and left the GBs while maintaining a detailed balance, so that the total number of atoms in

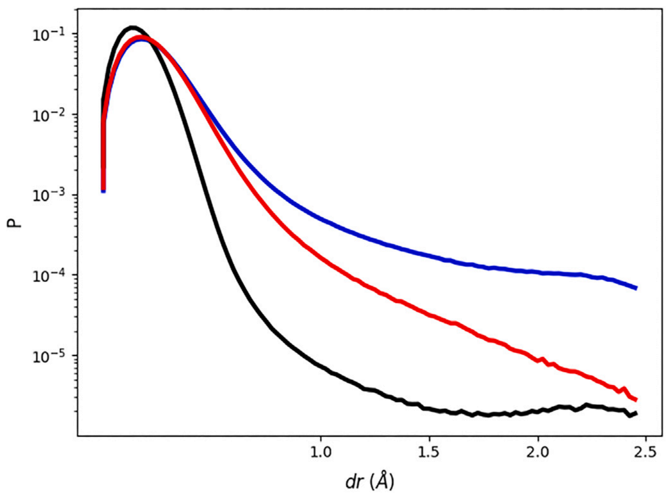

Fig. 2. Probability density of the displacement magnitudes of atoms in one cycle of IIC. The displacements were collected over 500 cycles in a simulation of 0.5% Cu-V, under a stress of 4 kbar. The lines, from bottom to top, are for the atoms within the interior of grains in an irradiated sample, atoms at GBs in a sample with no irradiation, and atoms at GBs in an irradiated sample.

the GBs remained approximately constant, see Fig. 3 for illustration. The flux was measured as the number of atoms in the group $\{GB_{flux}\} = \{GB_{post}\} - \{GB_{post}\} \cap \{GB_{pre}\}$, where $\{GB_{pre}\}$ and $\{GB_{post}\}$ are the groups of GB atoms before and after 500 cycles of the plastic stage of IIC. As seen in Fig. 4, which includes data from all the samples shown in Fig. 1, the creep compliance depends linearly on the flux of atoms. This is consistent with GB migration mediating plastic creep in the samples under study. However, counting the number of atoms within grains showed that grain coarsening did not contribute significantly to the observed IIC, see Appendix B. This suggests that shear deformation related to GB migration was the driving mechanism of the observed creep, in line with similar recent findings in nonirradiated samples [13–17]. We emphasize that this deformation occurs without net grain growth, highlighting that grain deformation, rather than grain coarsening, mediates the observed creep.

In polycrystalline materials the GB network is composed of GB planes and their intersections at triple junctions (TJs). The GB mobility is greater than that of the TJs, which exert a dragging force on the GB motion. TJ motion has a pronounced impact on the kinetic behavior of the GB network. Thus, controlling TJ mobility can be an important component in the reduction of GB plasticity [28–30]. In NCs, the volume fraction of TJs is greater due to the smaller size of the grains, and there is evidence that the TJ kinetics in NCs differ from those in polycrystals. Specifically, increased TJ mobility was observed, allowing stress-driven GB migration to occur without significant changes to GB curvature [13,31,32].

To differentiate between the relative contributions of the two mechanisms, GB and TJ mobility were analyzed and compared. This was done by identifying the GB and atom locations using the centrosymmetry parameter, see Appendix C, and then measuring their displacements. Note that since there is a flux of atoms in and out of the GBs, the GB displacement is not simply the sum of atom displacements within the GBs. Rather, we estimate the displacements of GB *locations* by identifying the closest points at which GB atoms were situated at the beginning and end of each cycle of IIC, $\Delta_i = \min_j \left| \boldsymbol{x}_{\text{post},i} - \boldsymbol{x}_{\text{pre},j} \right|$. Here, $x_{\text{post},i}$ is the coordinate of the ith atom in the GBs the end of the cycle, and $\boldsymbol{x}_{\text{pre},j}$ the coordinate of the jth atom in the GBs at the beginning of the cycle. This provides a minimal estimate of the extent of the local GB motion. The same analysis was repeated for TJ atom locations, to identify the specific contribution of TJ motion to the GB migration. The resulting distributions of displacements of GB and TJ locations are shown in Fig. 5. Displacements that are greater than the nearest-neighbor distance, $\Delta_i > 2.5$ Å, signify GB/TJ migration. It is seen that TJs are at least as mobile as GBs, as the probability of $\Delta_i > 2.5$ Å is greater for the TJ locations. Thus, the TJs do not exert a dragging force on the GBs, so that GB migration includes displacements of entire GB segments.

3.3. Driving mechanism of the GB migration

After establishing the mesoscopic mechanism of IIC in NC Cu-X alloys, we examine its microscopic origin. Dislocation extraction algorithm (DXA) [33] analyses show that there is a high density of dislocations at the GBs. Although GBs are obstacles for dislocation motion, dislocation climb in GBs is possible, and it has been shown to lead to creep [7,34]. To examine the role of dislocation climb in the samples under study, moving dislocations were monitored using the *common motion grouping* (CMG) method [35]. This method identifies groups of atoms that are displaced due to dislocation motion. The groups are found by organizing all atoms, whose displacement exceeds normal thermal Gaussian motion (r_{min}), as nodes of a binary graph. An edge connects every two atoms if (i) the distance between them is less than the next-nearest-neighbor distance in the GBs, and (ii) the cosine of the angle between their relative displacements has an absolute value that is greater than $\cos\theta_{\max} = \arctan(r_{\min}/b)$, where b is the Burgers vector length of a partial dislocation.

The cutoff for non-Gaussian displacement, $r_{\min}$, was estimated as 0.7 Å, based on the statistics of GB atom displacements. Correlative movement of atoms was identified as evidence of dislocation motion if the number of atoms participating was more than 4 [35].

Using this algorithm, we found that dislocation displacements occur only at GB areas. The average number of moving dislocations per cycle of IIC, normalized by the number of atoms in the GBs, is linear with creep compliance for all the samples studied in this work, see Fig. 6. The Pearson correlation coefficients of the number of moving dislocations and the creep compliance are 0.95, 0.94, and 0.8 for the Cu-V, Cu-Nb, and Cu-Ta samples, respectively, and the p-value under the null hypothesis is less than 6×10^{-6} in all three cases. This suggests that IIC is ultimately driven by dislocation dynamics at the microscopic level.

4. Mean-field model

Based on the observations that (i)GB migration acts as a mediator of IIC through shear deformation of the grains, (ii) the climbing of sessile dislocations in the GBs controls the GB migration rate, and (iii) the creep has a negligible thermal component [2,3], we propose a mean-field model of IIC.

Since the simulated IIC is controlled by GB processes, where climb is the dominant mechanism of dislocation motion [36], we calculate the strain rate due to climb only. This climb is due to the elastic interaction between the supersaturated irradiation-induced point-defect population and the long-range stress field of the dislocations. This type of interaction, known as *stress-induced preferential absorption* (SIPA), has been previously studied within the grains when intragrain processes dominate creep. In that case, a simple model was derived by describing the current of the point defects and their associated excess volume to dislocations. In this model, the current to dislocations with a Burgers vector aligned in direction j is assumed to be [1,37,38]

$$J_j = \rho_j \Omega \left(z_i^{dj} D_i C_i - z_v^{dj} D_v C_v \right). \tag{3}$$

Here, Ω is the atomic volume, and ρ is the uniform dislocation density within the grain. z_i^{dj} and z_v^{dj} are the capture efficiencies of interstitials and vacancies, respectively, by dislocations aligned in the direction j. $D_{i/v}$ and $C_{i/v}$ are the diffusion coefficients of the point defects and their concentrations. Assuming the simplest case, in which dislocations

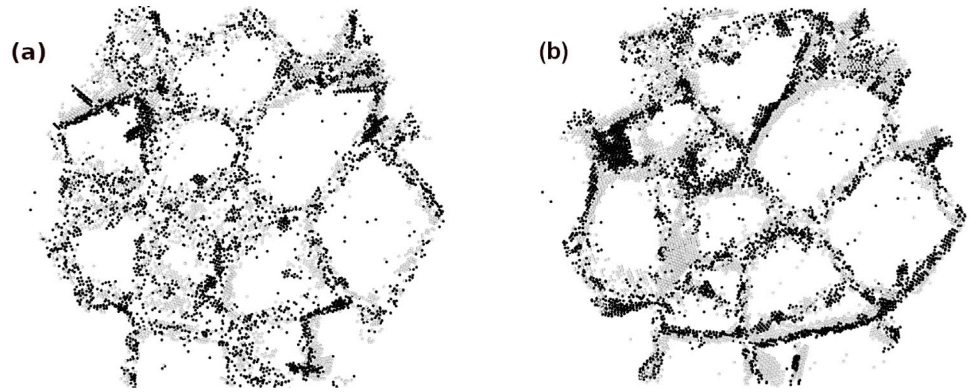

Fig. 3. A 1 nm-wide slab along the (111) plane of an irradiated (a) 5.3% and (b) 0.4% cubic Cu-V sample, 22 nm long, under a stress of 4 kbar. The black atoms are those that joined the GBs during 500 cycles of IIC, leading to an accumulated damage of ϕ = 0.4 dpa. The gray atoms are those that left the GBs during the same period of time.

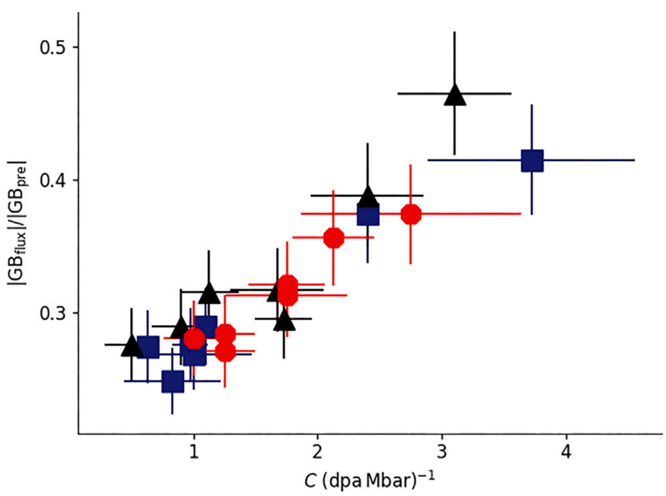

Fig. 4. Flux of atoms through the GBs as a function of creep compliance, during 500 cycles of IIC and under a stress of 4 kbar in NC Cu-V (squares), Cu-Nb (circles), and Cu-Ta (triangles), at varying concentrations.

are distributed isotropically among the three orthogonal directions, this leads to a strain rate of

$$\dot{\epsilon} = \frac{2}{9}\Omega\rho\left(\Delta Z_i^d D_i C_i - \Delta Z_v^d D_v C_v\right). \tag{4}$$

Here, $\Delta Z^d_{i/v}$ are the differences between the capture efficiencies of point defects by dislocations aligned and nonaligned with the applied stress. $\Delta Z^d_{i/v}$ were found to be proportional to the effective stress σ_e [1,37], assumed here to be identical to the applied stress.

When creep is dominated by GB processes, the creep resulting from defect–sink interactions at the GBs, where there is a high concentration of sinks, needs to be isolated. Due to this high concentration, the creep is dominated by defect–sink interactions, rather than defect diffusion within the GBs. Therefore, the defect introduction rates at the sinks $\rho\Omega D_{i/v}C_{i/v}$ in Eq. (4) are replaced by an expression proportional to the production rate of freely migrating point defects K_0, itself proportional to the dose rate $\dot{\phi}$ [1], so that $\dot{\epsilon} \propto \dot{\phi}\sigma_e$. Since in the NCs in this study the climb occurs *only in the GBs*, while the measured strain is that of the entire system, the strain rate $\dot{\epsilon}$ should be proportional also to the ratio between the volume of the GBs and the volume of the sample, itself proportional to $1/d$, where d is the average grain size [2,32]. Thus we have

$$\dot{\epsilon} = A\frac{\dot{\phi}\sigma_e}{d}, \tag{5}$$

with A a parameter depending on the structure of the crystal. If only point defects within a specific region near the GBs can reach them without recombining and influence the effective dpa, then $\dot{\phi}$ at the GBs is proportional to the global irradiation intensity $\dot{\Phi}$, and we have $\dot{\epsilon} \propto \dot{\Phi}\sigma_e$, in agreement with the findings of Refs. [2,3]. Deviations from this assumption depend on the interactions between GBs and defects, as well as between defects themselves, potentially introducing an additional dependence on grain size, up to d, in Eq. (5). The previous assumption, that the contribution of diffusion processes to creep within the GBs is small relative to that of sink–defect interactions, is consistent with the weak temperature dependence of the creep rate in the regime studied here.

In NC alloys with immiscible solute atoms segregated at the GBs a pinning stress derived from long-range dislocation–solute interaction, resulting from the size mismatch, reduces the effective stress σ_e on the dislocations [39–41], modifying the assumption that $\sigma_e = \sigma$ in Eq. (5). At low solute concentrations, the drag force is linearly proportional to the concentration [42], so that $\sigma_e = \sigma(1 - Bc_o)$, where σ is the external stress, B is a proportional constant, and the obstacle concentration

$$c_o = c_p + c_{fs} \tag{6}$$

is the sum of the precipitation concentration and the free solute atoms concentration, respectively. Therefore, the strain rate is

$$\dot{\epsilon} = A\frac{\dot{\phi}\sigma}{d}(1 - Bc_o). \tag{7}$$

Eq. (7) was fitted to the simulation results, with A held constant across the various systems, and B varied among the types of alloys (but held constant for all the concentrations in one type of alloy). This is because A is expected to depend on the solvent, while B is expected to depend on the size mismatch between the atoms of the solution, as well as their heat of mixing [43]. The fits are presented in Fig. 7(a–c), with $A = 31.4 \pm 2$ nm dpa^{-1} Mbar^{-1} and $B = 0.24 \pm 0.03$, 0.26 ± 0.03, and 0.3 ± 0.03, for Cu-V, Cu-Nb, and Cu-Ta, respectively.

The model was next compared to experimental measurements of IIC in Cu-W from Ref. [3]. In this experiment, thin films of $Cu_{93.5}W_{6.5}$ and $Cu_{99}W_1$, with an average grain size of 50 nm, were irradiated with 1.8 MeV Kr+ at a temperature of 473 K and a range of stresses of up to 10 MPa. TEM and STEM images taken before the irradiation indicated that W atoms were present both in the interior of the grains and at the GBs. After IIC, STEM images showed a large number of W precipitates, which did not exist prior to the irradiation. The diameter of the precipitates was up to 3 nm, and their densities in atom fraction were 2.4×10^{-5} and 5.9×10^{-6}.

To apply the mean-field model to the results of Ref. [3], which uses heavy ions to irradiate samples, two assumptions are made: First, it is assumed that only a small fraction of the defects produced during heavy ion irradiation escapes the cascade region and becomes freely migrating defects within the region near the GBs. Therefore, the ratio of the effective dpa in the simulation to the experimentally measured global dpa is equal to the *isolated point defect fraction* γ, which is between

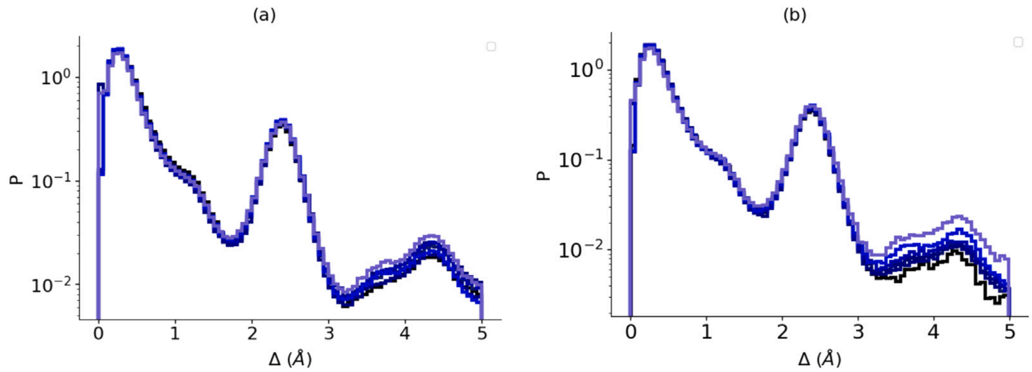

Fig. 5. Probability density of (a) GB and (b) TJ location displacement magnitudes, following one cycle of IIC, measured over 500 cycles of IIC in seven samples of Cu-V.

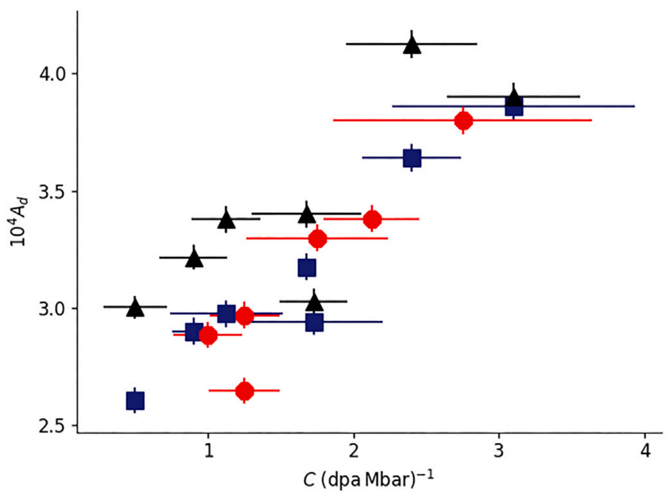

Fig. 6. The average number of moving dislocations per cycle, normalized by the number of atoms in the GBs, during 100 cycles of IIC and under a stress of 4 kbar in NC Cu-V (squares), Cu-Nb (circles), and Cu-Ta (triangles), at varying concentrations.

0.01 and 0.1 [1]. The creep compliance of the experiment was thus multiplied by $\gamma = 0.1$, to adjust it to the definition of Eq. (1) (as in Ref. [2]). Secondly, because the atom sizes of W and Ta are similar, and they both have a large heat of mixing in Cu, it is assumed that the fitted value of B for Cu-Ta can be applied to Cu-W.

Despite these approximations, Fig. 7(d) shows that there is good agreement between the creep rates as a function of the stress as measured in Ref. [3] and as calculated from Eq. (7). The creep compliance is the slope of the linear fits to the results. As mentioned above, only the densities of the W precipitates, c_p in Eq. (6), were given for the experiment. Thus, possible values for c_{fs} range from zero if all the solute precipitated, up to the atomic density of W in the alloy if only a small fraction of it precipitated, and the rest of it remained in the form of free atoms. This uncertainty is shown in Fig. 7(d) as the lower and upper bounds, respectively, of the fits for $Cu_{93.5}W_{6.5}$ and $Cu_{99}W_1$.

5. Conclusion

Irradiation-induced creep was simulated in nanocrystalline Cu-X alloys, where X is an immiscible solute atom with a large size mismatch and a positive heat of mixing. In these systems, the creep was shown to be strongly affected by the concentration of solute atoms and their segregation to the GBs, due to the effect of the solute on dislocation climb rates. The correlation between creep and climb rates was established using the recently-developed CMG method [35]. This correlation is attributed to the combined effects of GB migration and TJ mobility, leading to a stress-induced-preferential-absorption-based model of irradiation-induced creep in these systems.

The mechanism identified in this work, GB migration driven by dislocation climb, is well known in other contexts [7,14,16,17,32,44], but here it is shown to dominate IIC in NC materials. The simulations demonstrate that, under irradiation, point-defect absorption at GBs enhances dislocation climb, which in turn facilitates stress-driven GB migration. While GB sliding and grain coarsening are insignificant, grain deformation via GB and TJ migration significantly contributes to IIC.

A simplified mean-field model, extending previous work in large-grained systems, is fitted and shown to agree with recent experimental results. Establishing such a model may serve in the future to estimate the expected dependence of creep rates on different structural and chemical compositions. Thus, the model can contribute to the design of radiation-resistant materials, and in particular to the optimization of their structure and alloy compositions. Further study is needed to clarify the grain-size dependency of the model proposed here and its applicability to specific systems. However, the model can serve to estimate the lower limits of the applicability range of models fitted to experimental observations in larger grain-size materials.

CRediT authorship contribution statement

Noya Dimanstein Firman: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Eliyahu Zvi Engelberg:** Writing – review & editing, Writing – original draft, Software. **Yinon Ashkenazy:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Noya Dimanstein Firman reports financial support was provided by Israel Ministry of Innovation Science & Technology. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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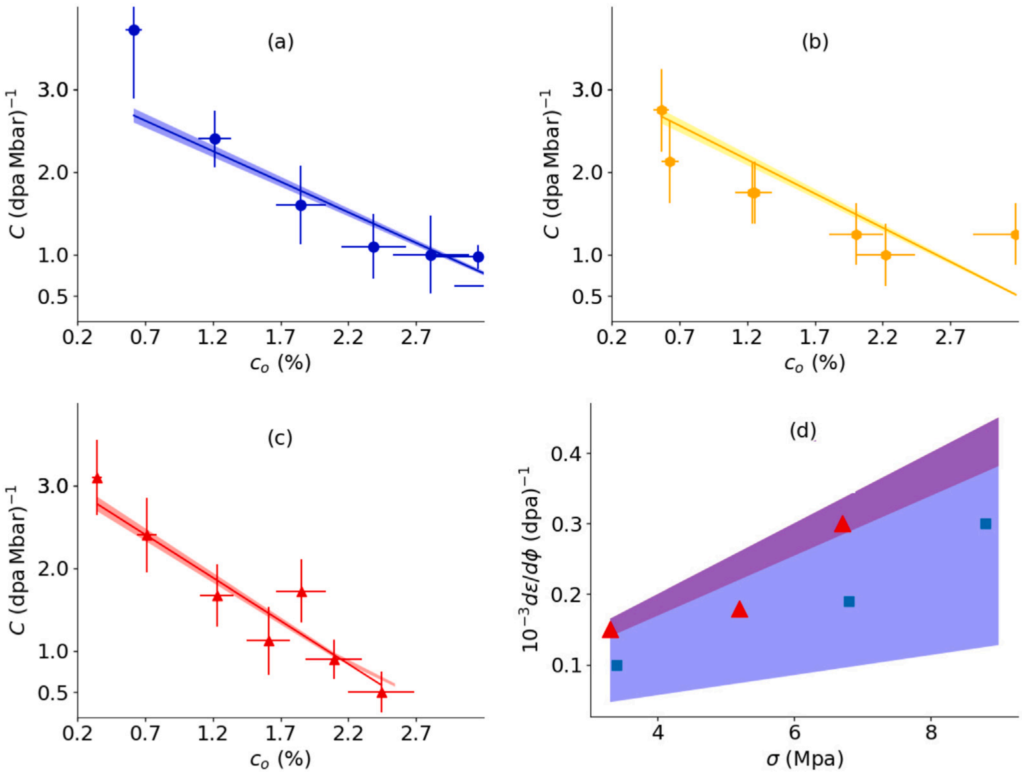

Fig. 7. Irradiation-induced creep compliance as a function of obstacle concentration at the GBs, in (a) Cu-V, (b) Cu-Nb, and (c) Cu-Ta, calculated from the simulation (points) and the mean-field model (lines). The error bars of the points are as in Fig. 1, and the shaded bands represent the confidence level of the parameter fit. (d) Strain rate divided by dpa as a function of the stress, measured in Ref. [3] in $Cu_{99}W_1$ (triangles) and $C_{93.5}W_{6.5}$ (squares), and calculated from the model for the same (top and bottom bands, respectively). The upper and lower bounds of the bands reflect the uncertainty in the degree of precipitation of the solutes, see text.

Appendix A. Calculating the creep compliance

Creep compliance quantifies susceptibility to creep. As mentioned in Section 3.1, it is the derivative of the sample strain ϵ by the dpa divided by the stress. In this section, we show how the strain and creep compliance are calculated.

The strain as a function of dpa is shown, for two different samples, in Fig. A.8. To find the creep compliance in the plastic stage of IIC, we examine the part of the graph that is approximately linear. ϵ was calculated as $\epsilon = (2dz - dx - dy)/(z_0 + x_0 + y_0)$, where x_0, y_0, and z_0 are the initial lengths of the sample, and dx, dy, and dz are the changes in lengths. For each sample, the slope of this part of the graph was divided by the applied stress of 4 kbar. Similar calculations were made for all samples in this study.

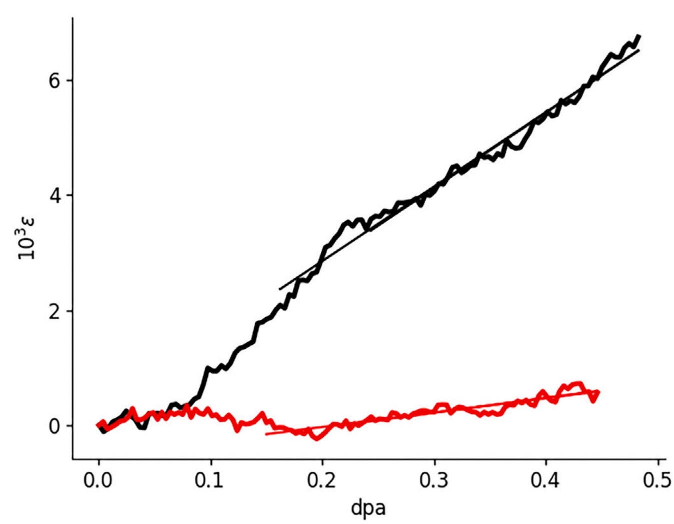

Fig. A.8. Strain as a function of dpa in 0.4% (top line) and 5.3% (bottom line) Cu-V samples subjected to IIC under a stress of 4 kbar. The slopes of the linear fits to ϵ(dpa) (straight lines) are the creep compliance in the respective samples.

Appendix B. Eliminating deformation mechanisms

To check for the occurrence of the sliding or gliding of GBs we monitored our samples for the sliding and gliding of grains, as it accompanies both processes [45]. This was done by dividing the simulation cell into 0.8 nm-thick planes, with 1.6 nm-wide gaps between each two planes, see Fig. B.9. The atoms in each plane were tracked over 500 cycles of the plastic stage of IIC. Grain rotation or sliding would change the shape of such a plane to a sawtooth (zigzag) pattern that breaks along GBs. Because the marked atoms stayed in the planes, sliding and gliding were ruled out. This test was repeated for planes oriented parallel to the xy, yz, and xz planes, for each of the samples examined in this work, yielding the same result.

To examine the possibility of grain coarsening occurring in the samples in the study, the number of atoms in each of the grains in each sample, before and after 500 cycles of IIC, were compared. The number of grains in each sample did not change. Furthermore, the difference in the number of atoms in each grain was within the margin of error. This is shown in Fig. B.10 for the samples of Cu-V. Similar results were obtained in Cu-Ta and Cu-Nb.

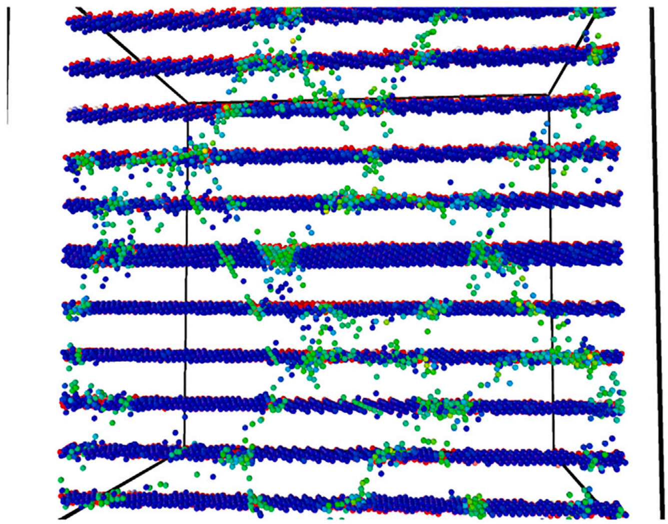

Fig. B.9. Atoms in 0.8 nm-thick planes oriented parallel to the xz plane in a cubic 0.4% Cu-V sample, 22 nm long, subjected to 500 cycles of IIC under a stress of 4 kbar. The initial and final positions of the atoms are colored red and blue, respectively. (The initial positions are mostly hidden behind the final positions.) The atoms in shades of green belong to GBs, and therefore were not taken into account when searching for slide and glide of the grains.

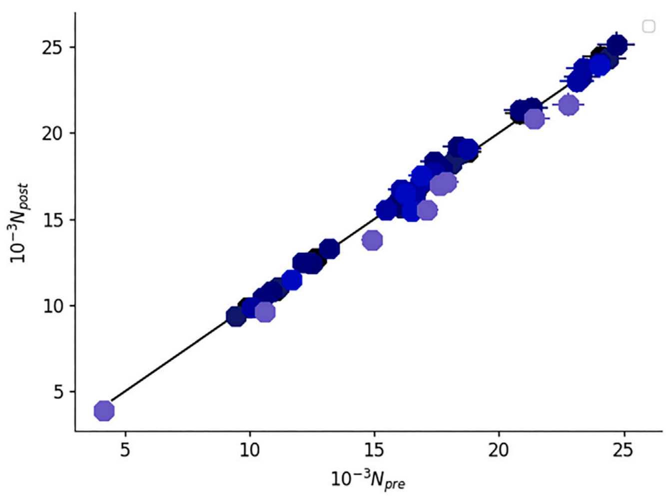

Fig. B.10. Number of atoms within grains of Cu-V samples at various solute concentrations, before and after 500 cycles of the plastic stage of IIC, under a stress of 4 kbar.

Appendix C. Identifying GB and TJ atoms

To determine which atoms in a sample belong to GBs, and which of those belong to TJs, each atom in the sample was identified as belonging either to a specific grain or to the GBs, using the centrosymmetry parameter. Then, if at least one of the atoms in a grain was at a distance of less than 1 nm from an atom at the GB, the GB atom was associated with that grain. GB atoms that were associated with more than two grains were identified as belonging to TJs. An example of this assignment in a small part of one sample is illustrated in Fig. C.11.

Data availability

Data will be made available on request.

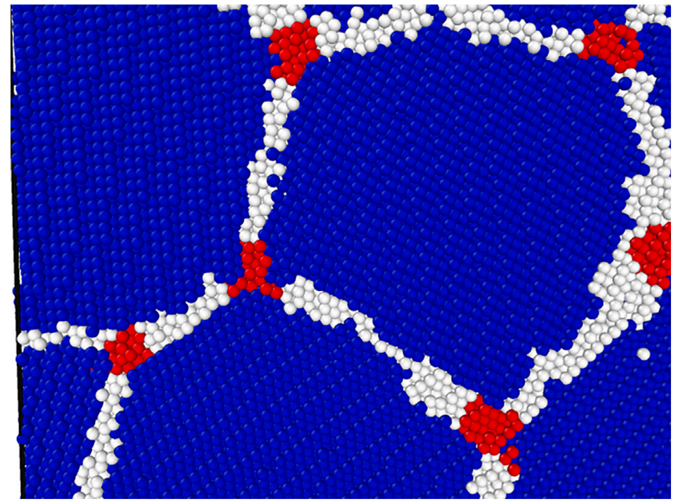

Fig. C.11. NC grain atoms (blue), GB atoms (white), and TJ atoms (red), in part of a 0.4% Cu-V sample.

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